

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Community Bridge Recommendation system

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>TriCore</i> <i>DSCPP-III-2026-T378</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Food is a basic necessities of life , yet a large amount of food is wasted everyday. A good surplus amount of food gets prepared daily at hotels , restaurants and events etc. . Whereas somewhere in the same locality or community or city there are organizations helping those in need of food. A meal that is left unused at a restaurant or a hotel or somewhere where its of could instead reach the need . This gap where the food is available without reaching the right place is difficult . The problem isn't only of finding someone to feed rather it's about reaching the right recipient . The organization could have different requirements , capacities etc. . The prepared or cooked food is also bounded by the restriction of time before it gets wasted which adds more challenge to the allocation of donation of the right recipient especially with multiple donors and recipients .

*Therefore the core issue becomes : **how can surplus food be allocated to a suitable recipient within its usable time .***

*Our project **Community bridge recommendation system** aims to address this surplus food redistribution and allocation problem by simply helping coordinators to identify feasible recipients and make better allocation decisions within the restricted time . Our recommendation system considers various relevant conditions from both donors (food type ,usable time , capacity etc) and recipients (availability , capacity etc) and then also ranks the recipients based on the inputs by respective users. Coordinators review the recommendations made by the system and then makes the final decision thus reducing the time taken to make decisions. The system also supports flexible allocation recommendations allowing even small number of meals reach where required . A transaction record is also made after the allocation is done keeping the record.*

Through the improvement in the way surplus food is allocated, the system can ensure that more usable food reaches those who need it instead of being wasted. It provides a structured way to make better use of surplus food while supporting the people and organizations involved in its redistribution.

State of the Art / Current solution (1 pt)

Currently, excess food redistribution is commonly coordinated by using NGO networks,volunteers, phone calls, messaging, spreadsheets, and manually planned pickup and distribution schedules.

A coordinator generally needs to identify a suitable recipient, check whether the organization is currently accepting food, determine whether it can handle the quantity and food type, and decide how quickly the donation needs to be moved. When a recipient cannot accept the complete donation, the remaining food may have to be handled separately.

Existing digital food-rescue platforms demonstrate that technology can support activities such as donation coordination, recipient matching, and multi-stop food rescue. However, our project focuses on creating a small, transparent academic prototype that demonstrates how fundamental data structures can support this decision-making process.

Since, our system is not attempting to build a production-scale logistics platform, it will have the essential decision workflow:

Donation → Feasibility → Recommendation → Human Decision → Allocation → Transaction Record.

Project Goals and Milestones (2 pts)

GOALS:

The primary goal is to develop a working decision-support prototype based on C++ so that it can organize surplus food redistribution using fundamental concepts of Data Structures and OOP.

The system will:

- Maintain donor and recipient information.*
- Allow coordinators to create and manage donations.*
- Evaluate recipient feasibility based on operational conditions.*
- Support both full and partial donation acceptance.*
- Prioritize donations according to usable time remaining.*
- Rank feasible recipients using deterministic criteria.*
- Allow the coordinator to make the final allocation decision.*
- Update donation and recipient states after allocation.*
- Maintain a persistent transaction history.*

MILESTONES

Milestone 1: Designing project models and basic C++ structure.

Milestone 2: To implement CSV-based donor and recipient data loading.

Milestone 3: To implement interactive donor, recipient, and donation management.

Milestone 4: To implement feasibility analysis and partial-match handling.

Milestone 5: To implement recipient recommendation using sorting.

Milestone 6: To implement donation urgency using a priority queue.

Milestone 7: To implement human-approved allocation and state updates.

Milestone 8: To implement transaction logging and persistence.

Milestone 9: Performing integrated testing, including the 250 → 140 partial-allocation scenario.

Future milestone: Will introduce distance/travel-time consideration into feasibility and recommendation.

Project Approach (3 pts)

The project will implement as a C++ console-based application by using object-oriented design and some data structures concepts.

This system will be maintaining Donor, Recipient, Donation, and Transaction objects by using a central DataManager. Persistent information will be stored in local CSV files using C++ file streams. The application is following a decision-support workflow. When a donation is created, the system will first evaluate potential recipients using feasibility conditions like current acceptance status, food compatibility, available capacity, and pickup capability.

Feasible recipients are then ranked using `std::sort`. The current recommendation will be prioritizing::

- 1. Higher potential food acceptance.*
- 2. Smaller unused capacity when acceptance is equal.*
- 3. Recipient ID as a deterministic tie-breaker.*

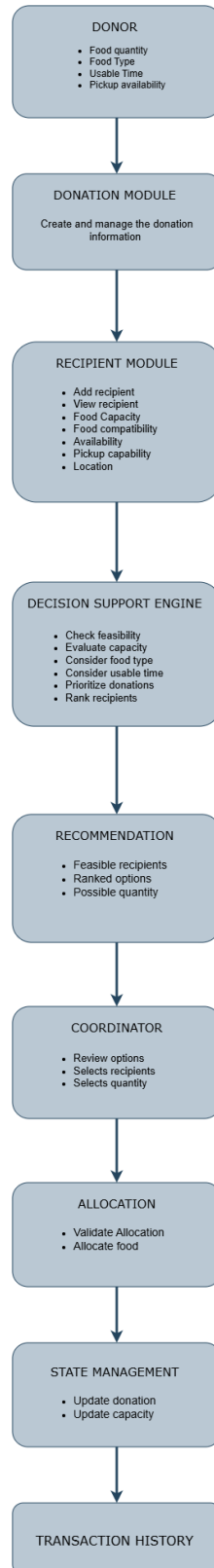
Pending donations will be managed using a `std::priority_queue`, where donations with less usable time remaining will receive higher priority.

The final allocation is human-in-the-loop. The coordinator will select the recipient and quantity, then the system validate the decision and will update both donation and recipient states.

Successful allocations will be stored as transactions and persisted in a CSV file.

As a future enhancement, donor and recipient locations can be represented using coordinates so that the estimated distance/travel time can be used for feasibility and recommendation.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The Community Bridge Recommendation System is developed to support and improve the decision made by the coordinator to ensure that the surplus food reaches to those in need. The complete system will allow a coordinator to create donors, recipients, donations, modify details and with the help of recommendation system the process of allocation would be more efficient as it is capable of handling partial food allocation too. After a successful donation allocation the system also keeps a consistent donation and recipient state and generate transaction record. This project is a demonstration of how fundamental Data Structures and C++ concepts such as vectors, linear searching, sorting, priority queues and file handling etc. used together to solve the problem of efficient food redistribution.

Assumptions

This prototype will assume that donor and recipient records are predefined and maintained locally using CSV files. Recipient capacity will represent currently available capacity. Food compatibility and pickup capability will be represented using predefined attributes. Donation usable time will be entered in minutes. The system will not use real-time GPS, traffic, external routing APIs, databases, or machine learning. Final allocation decisions will be made by a human.

References

1. **ISO C++ Standard Library Documentation** — Reference for `std::vector`, `std::sort`, `std::priority_queue`, strings, and file streams.
2. **C++ reference — C++ Standard Library** — Reference for C++ containers, algorithms, file streams, and standard library operations.
3. **Data Structures and Algorithms Course Material** — Reference for searching, sorting, priority queues, complexity analysis, and related DSA concepts.
4. **Object-Oriented Programming with C++ Course Material** — Reference for classes, objects, encapsulation, constructors, and member functions.
5. **Project-specific design and testing documentation** — Used to define the Food Rescue workflow, feasibility rules, recommendation process, allocation logic, and transaction handling.